

Project Plan

Improving Declustering for Qcify Air



Overview

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- QCIFY AIR MACHINE OVERVIEW
- THE PROBLEM
- WHY DECLUSTERING MATTERS
- POSSIBLE CONSTRAINTS
- METHODOLOGY
- TIMELINE
- RESULTS & IMPACT
- COMMUNICATION & PROGRESS REPORTING
- EXPECTED DELIVERABLES

About



member of  **eivis**



Qcify is an innovative company that creates automated quality control solutions for the food industry.



Its systems make inspection faster, more reliable, and more data-driven.



Based in California and Belgium, Qcify is growing internationally and supports customers around the world.

Qcify Air Machine Overview

- AI -powered in-line optical sorter
- Designed for grading nuts
- Real-time adaptive classification
- Precise defect removal



The Problem

**Products overlap or
form clusters on the
conveyor**

Innaccurate separation leads to:

- Reduced detection accuracy
- Lower classification quality
- Decreased sorting performance

Why is it important?



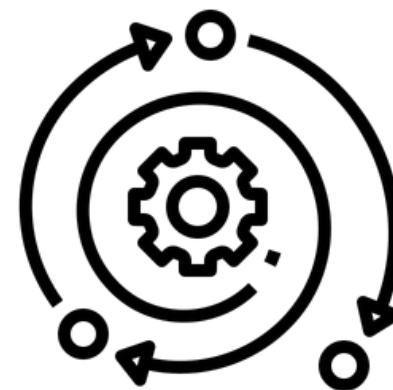
Accuracy

Better product separation leads to more accurate detection



Throughput

Efficient declustering supports high-speed line performance



Business Impact

More reliable inspection creates value for both production and quality teams

Methodology (R&D Focus)

1

Research possible solutions

2

Implementation and testing

3

Evaluation & Comparison

Important Requirements

REQUIREMENT	WHY IT MATTERS
Real-time implementation	Cannot increase latency
Pipeline compatibility	Must integrate with current system
Variability of products	Needs to generalize

Timeline

Week 1-3

System understanding and research

Week 4-10

Testing, validating and code implementation

Week 4-6

Implementing and testing first approach: 3-class semantic segmentation (added extra class: nut boundaries)

Week 7-9

Implementing and testing second approach: distance maps + nut boundaries

Week 10

Implementing and testing third approach: Training with settings above, but on different model backbone

Week 11-13

Validation, documentation and final improvements

Expected Results

- Overview of approaches and techniques explored
- Identification of main challenges and limitations
- Ideally, a working solution for improvement
- Potential integration of the best-performing approach into existing pipeline

Communication & Progress Reporting



Regular mentor meetings

1-2 meetings per weeks to discuss progress, challenges and next steps



Weekly status report

Sent to internship coordinator summarizing progress



Project tracking

Tasks and development progress managed using Azure DevOps

Deliverables

Project Plan

Realization (Thesis Work)

**Documentation Update
(Company Wiki)**

Code Base/Implementation

Reflection & Recommendations